Welcome to CSS 599

Autumn 2025

Graduate Research Seminar

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Syllabus Review – See Canvas Site: CSS 599 A Au 25: Faculty Research Seminar

Understanding Research: Some core concepts and things to think about when listening to presentations

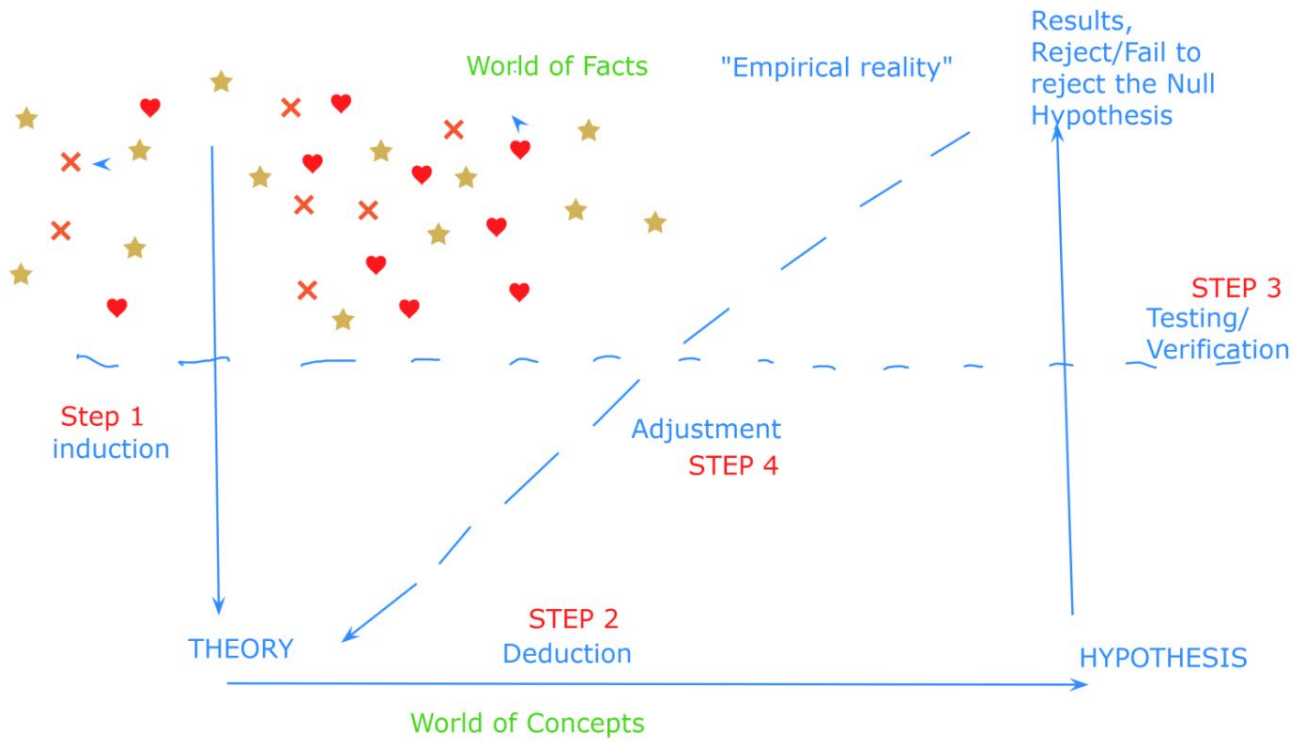
Or, better titled as …

“Things I wish I knew about listening to research when I started grad school” 😊

A definition of “science” → What (good) researchers do. 😊

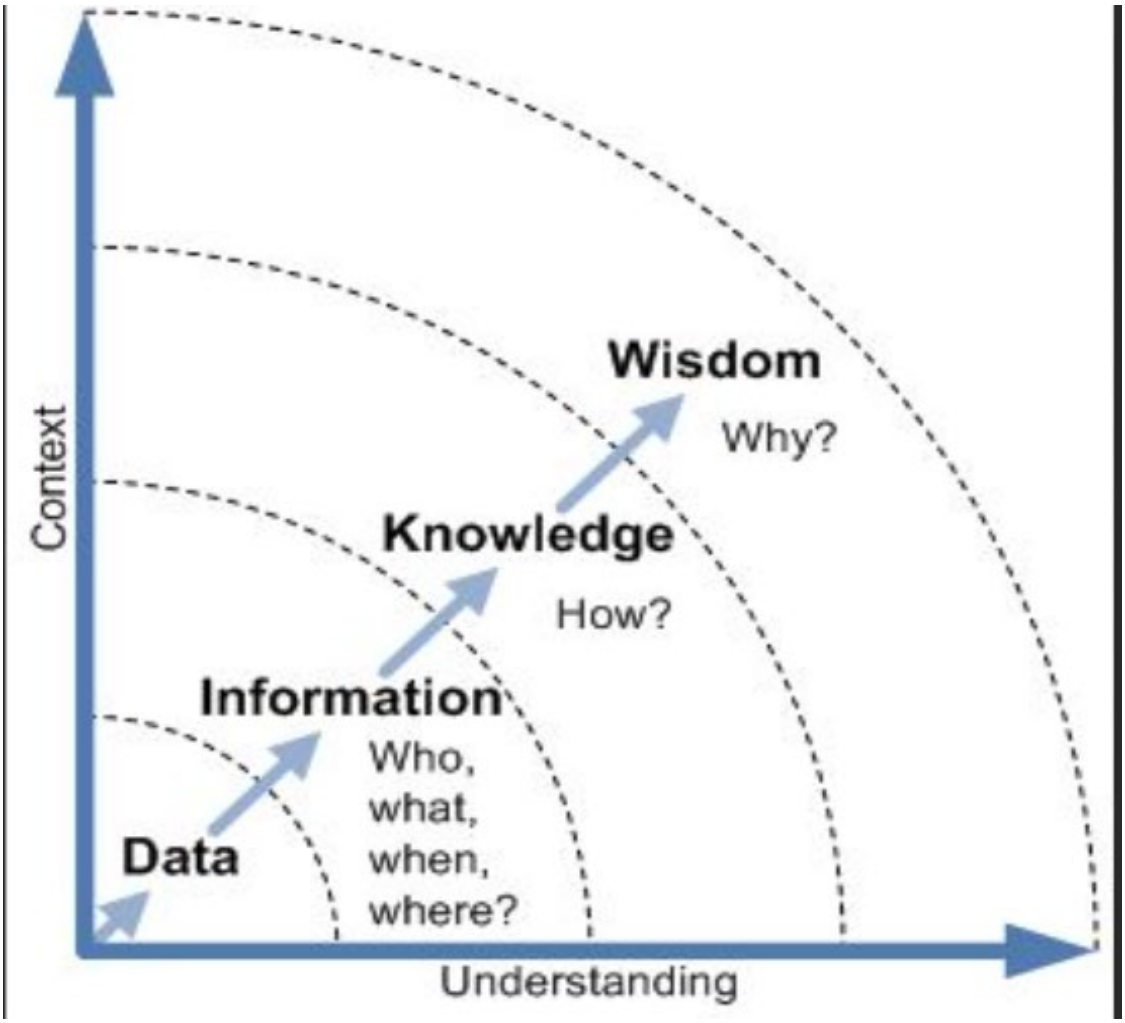
Science is the systematic study of empirical phenomena.

The Empirical Model – An Overview of the Iterative steps



DIKW Model: [DIKW pyramid - Wikipedia]

DATA →INFORMATION → knowledge → wisdom → (and maybe) “truth”



There are many methods for doing research. An overview follows:

> *(Hint: Take CSS 565: Research Methods in software Development for a deep level understanding of these approaches – and much more about research/metrics/processes.)* 😊

NATURALISTIC OBSERVATION

- UNOBTRUSIVE MEASURES/METRICS – observing w/o impacting the subjects being studied, Hawthorne studies
- CASE STUDIES – DEPTH, LOT OF IDEAS FOR HYPOTHESES, FOR INDUCTION
- ETHNOGRAHPIC STUDIES

CORRELATIONAL STUDIES -- Where "r" is the correlation coefficient, ranges from -1 to 0 to +1;

1) Correlation examines the overlap or "shared variance" between variables.
2) "Correlation does not imply causation"

TRUE EXPERIMENTAL DESIGNS a.k.a. CLASSICAL RESEARCH/LAB STUDIES

"A/B" testing – or even "A → Z testing," (example of testing different features in large, widely played games – e.g., testing the impact of choice of the color of a tractor in Farmville; testing variations in gameplay for "franchise-level" games, regional studies)

QUASI-EXPERIMENTAL DESIGNS – NOT RANDOM ASSIGNMENT OF SUBJECTS TO IV (Independent variable) or CONDITIONS (or levels)

> Sometimes you cannot assign treatments/conditions to subjects such as people to specific groups/settings; demographic and psychographic variables that cannot be randomly assigned, health conditions, etc.

COMPUTER EXPERIMENTS

- SIMULATIONS/MODELLING

- Machine Learning/Artificial Intelligence /Chatbots

    - Robotics
    - Sensor Arrays
    - Predictive analytics
    - An infinite list …

SEMANTIC WEB ANALYTICS – Extracting meaning from data via the use of tools such as RDF, SPARQL – and generative AI techniques

> *RDF (Resource Description Framework), SPARQL (protocol and query language) – useful tools for exploring semantic web concepts; extracting meaning/knowledge from various statements.*

SURVEYS/QUESTIONNAIRES – Many forms and types of surveys – and modern methods for collecting data from a wide variety of sources.

FOCUS GROUPS – structured interviews versus unstructured;

> Core questions -- what do you like best about X; what do you like least about X, if you had one wish related to X, what would it be -->"”" is called the attitude object “object/concept of interest”

SOCIAL NETWORK ANALYTICS/SOCIAL COMPUTING

> A way to evaluate how members, cliques impact one another – how do networks change, how do they adapt to new situations, ideas, feedback…; persuasive technologies

META-ANALYSIS “analysis of analyses” – study of studies → comparing a set of research paper/etc. across a wide range of shared characteristics or attributes. Things that a meta-analysis might address include:

Attributes and differences across studies – for example:

1) Criteria for selecting the studies to be included in the meta-analysis
2) number of citations → implies relevance, usefulness, originality, what number of people are doing similar work that might be leveraging what you have found – basis for a next iteration of the idea

3) Statistical findings – p-value, statistical significance of the work; what statistical techniques were used to analyze the data; results
4) When was the study conducted?
5) Common keywords – shared vocabulary across studies
6) Methodologies used -- maybe different methods produce different results; or maybe the same/similar -- > builds confidence due to examining the hypothesis from many different angles
7) Interpretation of the results/findings → perhaps create a rubric for scoring, classifying, evaluating…
8) Subjects used – who or what was studied; the situation the objects of study were in at the time; setting
9) Characteristics of the sample being studied – size of the training data set, quality of the data….

When listening to research, you should think about limitations/potential sources of problems with research results such as:

Threats to internal validity -- things that occur/processes that might bias and/or influence the outcome in a systematic way that may not reflect reality.

Example: Effects of training on programmer productivity. What alternative influences can make it look like training impacts productivity?

Threats to external validity (generalizability of our findings/results to the real world). Can the results/findings for your study be used/applied outside of the research environment? Is the application of the research/ability to achieve certain results constrained by the manner in which the research was conducted?

Some other critical topics/questions to think about as you select a research topic/project/group:

1) *When choosing a research group and/or mentor, what critical questions/issues must be addressed to ensure a quality experience for a graduate student?*

2) *What are requirements for how we treat and manage and archive data (e.g., NSF, government grants require that the data be stored, saved, and shareable so that other researchers can look at it – of course, anonymity/confidentiality requirements must be maintained)?*

3) *What are examples of good advising/mentoring techniques? What works for you?*

4) *Ethics. What is a mistake? Negligence (haste, carelessness, inattention, laziness…) as a rationale for a mistake?*

5) *What is meant by research misconduct – fabrication, falsification; what is the purpose of an IRB – and its role and responsibilities? What is "Common Rule?" What is meant by informed consent and anonymity of response (no identifying info can be shared for your subjects)? Role of*

social media, analytics, "scraping" from applications/web sites – is a EULA the answer?). Role of deception?

6) *What is the purpose of peer review – and the processes related to publishing articles that help ensure rigor and quality? How is authorship/allocation of credit determined? How can you determine whether an article is published in a "refereed journal?"*
    a. Conferences
    b. Workshops
    c. "top tier" academic journals. ACM, IEEE – also important as your work may be cited by others; technical white papers
    d. Usually, the order of authorship is 1st – idea creator/and worker; level of contribution is next; some fields just do alphabetical order; some fields add everyone who even said the keywords as they relate to your paper – very inclusive; what you should do is discuss order of authorship as you begin to determine the actual research study and who will do what?
    e. Sharing knowledge is the primary objective…should determine whether the knowledge is proprietary, NDA – suggest looking up some sample NDAs on the web
7) *What is meant by intellectual property, conflict of interest and other issues related to funding sources for research? Purpose of an LLC and protections provided for companies?*
8) *What should you do if you have an idea that has commercial value/possibilities that may be part of your schoolwork?*

A Useful Reference about general concepts related to doing research:

On being a scientist: A guide to responsible conduct in research. (2009). National Academy of Sciences, National Academy of Engineering, and Institute of Medicine. (Third Edition) See: On Being a Scientist: A Guide to Responsible Conduct in Research - PubMed (nih.gov)

FINALLY – AND MOST IMPORTANT:

- LEARN ABOUT WORK THAT IS BEING DONE OUTSIDE OF YOUR COMFORT LEVEL
- LISTEN CAREFULLY AND RESPECTFULLY
- ASK THOUGHTFUL QUESTIONS
- ENJOY THE OPPORTUNITY TO LISTEN TO PEOPLE AS THEY TALK ABOUT THEIR RESEARCH PASSION/POTENTIAL FOR IMPACT
- BE AN ACTIVE MEMBER OF THE RESEARCH COMMUNITY
- SHARE YOUR WORK THROUGH PRESENTATIONS, PUBLICATIONS, CONFERENCES -- AND WITH YOUR FRIENDS!

And, HAVE FUN!